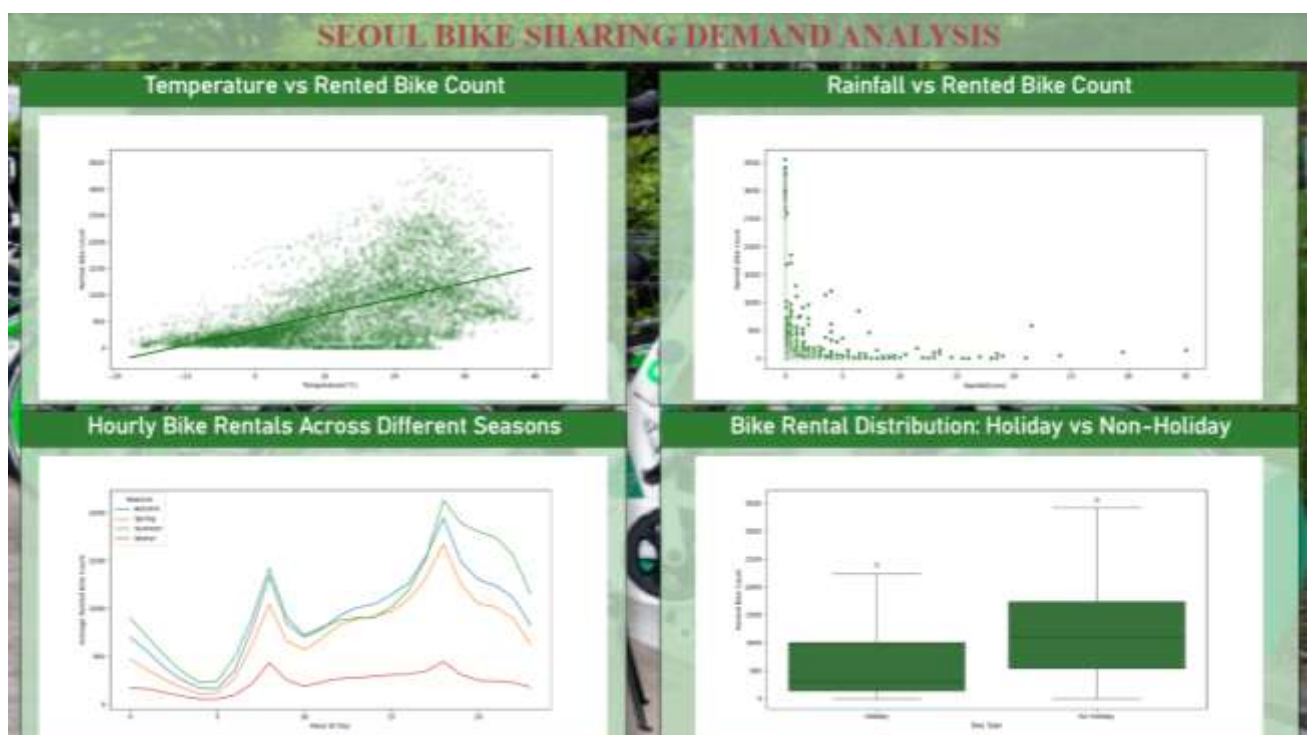
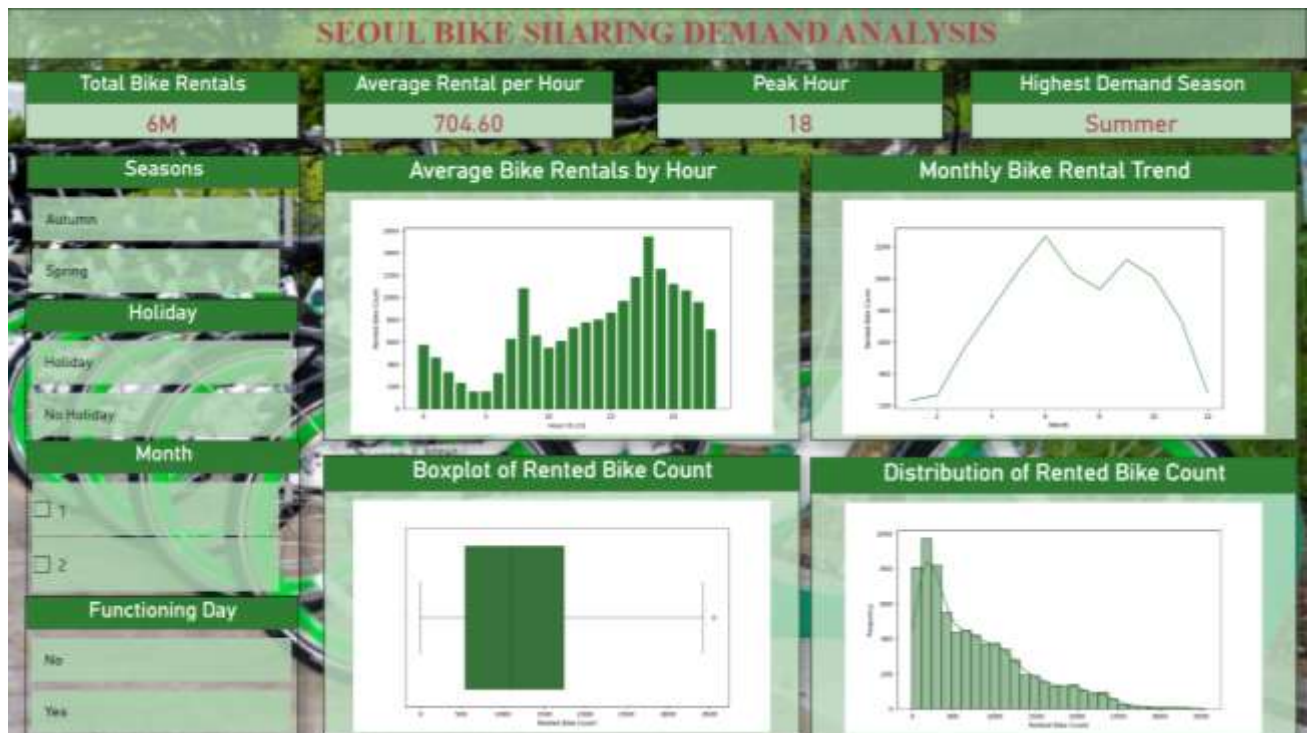
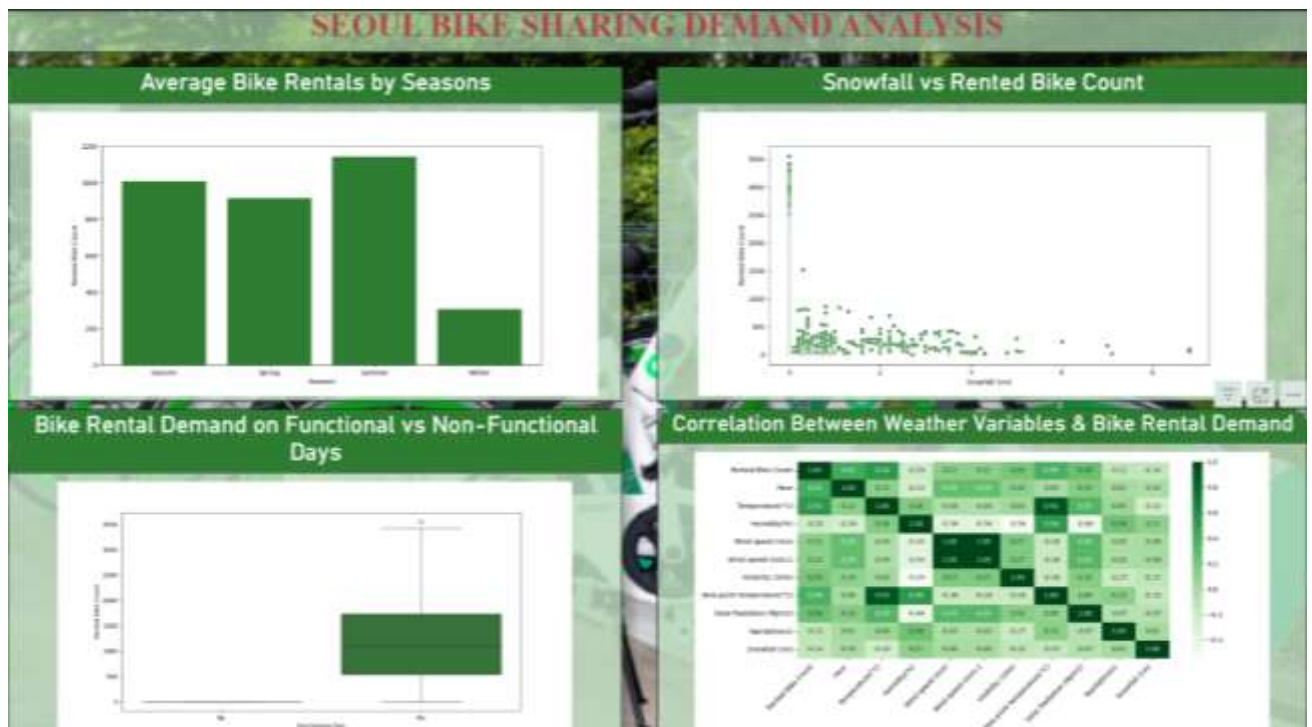


SEOUL BIKE SHARING DEMAND ANALYSIS

Power BI Dashboard





The Power BI dashboard gives a complete view of bike rental demand patterns in Seoul by combining time-based trends, weather conditions, seasonal effects and operational factors.

- The first dashboard shows the overall demand trends by hour, month, season and holiday.
- The second and the third dashboard focus on weather impact, seasonal variation and system availability on bike rental.

Findings Summary:

- Bike rental demand in Seoul is not uniform and changes significantly across seasons, hours and weather conditions.
- Rentals demand peak during the mornings and evenings, showing a strong commuter usage.
- Warm and dry weather conditions increases bike usage, whereas rainfall and snowfall reduce demand.
- The usage of bikes peaks during summer and reduces during winter indicating seasonal patterns.
- Bike rentals are mainly driven by functioning days and non-holidays, highlighting the commuting behaviour rather than for leisure reasons.

Key Findings:

- Bike rentals peak during office commute hours, indicating that bikes are mainly used for daily transportation.
- Weather is the most influencing external factor, with temperature having a positive correlation with bike rentals, while rainfall and snowfall have a negative impact.

- Seasonal demand is highest in summer, while it drastically reduces during winter due to unfavourable weather conditions.
- Non-holidays have higher demand, proving bikes are used more for routine travel.
- Rentals drop to almost zero on non-functioning days, indicating that system availability directly influences the bike demand.

Business Recommendation:

1. Resource Optimization

Bike availability shall be increased during peak commute hours and summer months, and reduce supply during low-demand period.

2. Weather-Based Planning

The bike distribution and maintenance schedules can be adjusted based on weather forecasts, especially during heavy rain or snow.

3. Seasonal Maintenance

The major maintenance and repairs shall be planned during low-demand period like winter months.

4. Commuter-focused Policies

Weekly commuter passes or subscriptions can be introduced to target primary user base and improve customer retention.

5. System Reliability

Ensure maximum system uptime on working days, as even short periods of downtime can lead to a greater loss in demand.

Conclusion:

The dashboard clearly shows how bike rental demand works in Seoul.

Bike usage is highest during **morning and evening office hours** and increases in **warm, dry weather**.

Working days have much higher demand than holidays, showing that rentals are mainly driven by daily commuting.

Seasonal trends indicate **peak usage in summer** and **lowest demand in winter** due to weather conditions.

Overall, these insights help service providers **optimize bike availability, plan maintenance**, and support better **urban transportation planning**.